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Artemisinin ameliorates intestinal inflammation by skewing macrophages to the M2 phenotype and inhibiting epithelial–mesenchymal transition

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A R T I C L E I N F O

Keywords:
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A B S T R A C T

Background: Inflammatory bowel disease (IBD) is a self-destructive intestinal disease whose etiology is unclear but complex and the effective treatment is deficient. Increasing evidences have indicated that immune dysfunction and epithelial–mesenchymal transition (EMT)-related intestinal mucosal barrier impaired hold critical position in the pathogenesis of IBD. Artemisinin (ART) is a sesquiterpenoid compound extracted from Chinese herbal medicine which has good immunomodulatory effects. Studies have shown that artemisinin and its analogues have therapeutic effects on a variety of tumors and immune-related disorders. The purpose of current study was to research the effect and mechanism about artemisinin-induced macrophage polarization to M2 phenotype and inhibiting the process of EMT.
Methods: In vitro, the anti-inflammatory effect of artemisinin is mainly verified by RAW264.7 cells and tissue (colon tissue and PBMC) from CD patients with active intestinal inflammation. RAW264.7 cells stimulated with LPS to induce inflammatory state and ART were used as therapeutic treatment in different concentration. Then the expression levels of pro-inflammatory factors, macrophage polarization and ERK pathway were analyzed. Colon tissue and PBMC from CD patients were treated with ART in different concentrations and macrophage polarization, pro-inflammatory factors expression, EMT-related protein were analyzed. In vivo, DSS-induced colitis mice were treated by ART for seven days. The DAI score was calculated and the colons and spleens were harvested after the animals were sacrificed. The expression of macrophage markers and EMT-related markers in the intestines of mice in each group were monitored by qPCR and western blot.
Result: ART treatment could decrease the levels of pro-inflammatory coefficient expressed in theRAW264.7 cells and human PBMC. Moreover, ART could ameliorate the intestinal inflammation *in vivo* through down-regulating the expression of pro-inflammatory factors, promoting macrophage polarization to M2 phenotype and inhibiting the process of EMT.
Conclusion: Taken together, our findings demonstrated that artemisinin might ameliorate inflammation by inducing macrophage polarization to M2 phenotype and inhibiting the process of EMT, suggesting that ART may be applied to the rehabilitation of IBD in the future.

1. Introduction

Inflammatory bowel disease (IBD) is a kind of immune related disease with unclear complex pathogenesis [1–3] which can be mainly divided into ulcerative colitis (UC), meanwhile include Crohn's disease (CD) by clinical manifestation and endoscopic performance. The incidence of IBD rapidly increases globally [4] and likely to reduce the quality of life of patients with IBD significantly. Moreover, the medical expenses, mortality and disability caused by IBD also impose a dramatic burden on social and economic development [5].

More and more researches have indicated that the role of the gastrointestinal mucosal in maintaining biological immune homeostasis cannot be ignored [6]. As a central part of innate immunity, macrophages are differentiated into a variety of phenotypes [7] and carry out a range of functions through interacting with antigen from food and intestinal microorganism. Among the various phenotypes of macrophages, M1 phenotype and M2 phenotype perform opposite functions in inflammation regulation. M1 phenotype promotes inflammatory process

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[8], damages surrounding tissue and inhibits the repair and regeneration [9] by secreting inflammatory cytokines (TNF-α, etc.), including iNOS, TNF-α, IL-6, CD86, etc. [10] On the contrary, the major functions of M2 phenotype are manifested as immune tolerance, which includes inhibiting inflammation, tissue repair, angiogenesis, etc. and rely on the production of CD206, Arg1, etc.[8] The imbalance of classical M1 and alternative M2 is always associated with the development of inflammatory diseases [11–13], and inducing differentiation of macrophages into M2 may be a effective solution to treat these clinical disorders.

Chronic inflammation associated with IBD can lead to epithelial-mesenchymal transition (EMT), that is, intestinal epithelial cells lose their characteristics of epithelial cells and in turn acquire characteristics of mesenchymal cells. In this process, the expression of various tight junction proteins as markers of epithelial cells, such as zona occludens (ZO)-1, occludin, is down-regulated, while the protein markers associated with mesenchymal cells (aSMA, vimentin, fibronectin, and collagen) and the transcription factors of EMT (Twist, Snail, Slug, Zeb1/2) were up-regulated [14]. The occurrence of EMT may be have relevancy with the massive expression of pro-inflammatory cytokines caused by intestinal immune disorders [15,16]. The results of the studies have confirmed that EMT plays a pivotal role in CD-related fistula [17] and IBD-related intestinal fibrosis [18]. Therefore, inhibition of the development of EMT could be a feasible method for the rehabilitation of IBD.

Mitogen-activated protein kinase (MAPK) is proved to be important for the regulation in cell differentiation, division, and apoptosis [19]. Extracellular signal-regulated kinase 1/2 (ERK1/2) is one of the most studied MAPK family members and have closely linked to the pathogenesis of IBD [20,21]. The activation of ERK pathway could cause the secretion of pro-inflammatory cytokines and result in abnormal level of intestinal inflammation. Therefore, inhibition of ERK pathway activation was determined as an anti-inflammatory process to against IBD.

Artemisinin (ART), the active composition extracted from leaves of Artemisia annua L, was first found in 1972 by Youyou Tu, which is widely known for its antimalarial activity. In the past decades, researches on artemisinin and its derivatives have been extended to the field of tumors and adaptive immunity-related autoimmune diseases [22–24], which provides the rationale that ART might be used for the treatment of IBD. However, the protective effects of ART on IBD keep uncertain. From the latest research process, we used RAW264.7 cell line and DSS-induced experiment colitis mice model to evaluate the protective effects of ART and to verify whether artemisinin can induce macrophages differentiation into M2 phenotype and regulate the process of EMT. Furthermore, we also used PBMC from IBD patients to confirm the role of artemisinin in the division of macrophages in vitro.

2. Method and material

2.1. Cell culture

RAW 264.7 cells were obtained from American Type Culture Collection (ATCC, Manassas, VA, USA). The cells had cultured in DMEM (Gibco, Life) which with 10% bovine serum (Gibco, Life), 100 U/mL penicillin meanwhile include 100 μg/mL streptomycin. In this study, cells were cultivated in the humidified incubator with 5% CO_2 at 37 °C. The cells were stimulated with LPS (1 μg/mL) and treated with different concentration of ART.

2.2. Mice

Female C57/BL6 mice (6–8 weeks, 19–22 g), which were obtained from Shanghai Super—B&K laboratory animal Corp. Ltd (Certificate No 2018-0006). Then put all mice in specific pathogen-free (SPF) environment. 32 mice were separated into different four groups randomly: include normal control group, vehicle control group and ART (20 or 80 mg/kg) group. The administration of 4% (*w/v*) DSS in drinking water was used to induced experiment colitis was induced. Mice got DSS distilled water (model) or regular drinking water (normal control) for 7 days, then continue this study for another 7 days and treated with ART or vehicle orally as described previously [25]. The disease active index (DAI) was calculated every day [26] as previously described. All mice were died at the 15th day and their spleens and entire colons were harvested for further experiments.

2.3. RT-PCR

Total RNA was preserved in RAW264.7 cells, and we served GAPDH as the internal control. RNA was reverse-transcribed into cDNA using reverse transcriptase. Then put PCR into final volume of 10 μL, and use a PCR reagent kit (Takara, Japan). The primer sequences used for qPCR analysis were showed as follows: mouse *GAPDH*, forward, 5′-AACTTTGGCATTGTGGAAGG-3′, and reverse, 5′-CACATTGGGGGTAGGAACAC-3′; mouse IL-1β, forward, 5′-ACGGACCCCAAAAGATGAAG-3′, and reverse, 5′-TTCTCCACAGCCACAATGAG-3′; mouse IL-6, forward, 5′- GATAAGCTGGAGTCACAGAAGG-3′, and reverse, 5′-GGAATGTCCACAAACTGATATGC −3′; mouse TNF-α, forward, 5′-CAGGCGGTGCCTATGTCTC-3′, and reverse, 5′-CGATCACCCCGAAGTTCAGTAG-3′; mouse Snail, forward, 5′-CACTGCCACAGGCCGTATC-3′, and reverse, 5′-CTTGCCGCACACCTTACAG-3′mouse Twist, 5′- CGCTACAGCAAGAAATCGAGC-3′, and reverse, 5′-GCTGAGCTTGTCAGAGGGG-3′; mouse Vimentin, forward, 5′-TTTCTCTGCCTCTGCCAAC-3′, and reverse, 5′-TCTCATTGATCACCTGTCCATC-3′; mouse Fibronectin, forward, 5′-CTTTGGCAGTGGTCATTTCAG-3′, and reverse, 5′-ATTCTCCCTTTCCATTCCCG-3′; mouse ZO-1, forward, 5′-GGTCCCGTCAGAGGTCTATTC-3′, and reverse, 5′- GTCTGTGAGGAGCTTCGTTTG-3′; mouse Claudin, forward, 5′- ATGGGCTTAGTCTTCCGAACG-3′, and reverse, 5′- CTTCCAGTGCGGCAAGTAGTT-3′; mouse Occludin, forward, 5′- TGAAAGTCCACCTCCTTACAGA −3′, and reverse, 5′- CCGGATAAAAAGAGTACGCTGG-3′; human GAPDH, forward, 5′- TGTGGGCATCAATGGATTTGG-3′, and reverse 5′-ACACCATGTATTCCGGGTCAAT-3′; human IL-1β, forward, 5′-GAGCCCCAAAAGCAAGAGGAA-3′, and reverse 5′-TGCGGGCATACGGTTTCATC-3′; human IL-6, forward, 5′-CTCAACGTGGCATAAGACTACAT-3′,and reverse, 5′-CCTCCTTGTATCCAGCCATTCT-3′; human TNF-α, forward, 5′- AGGACGACTGTTCAGCACG-3′, and reverse, 5′- CCGGGCAACAATGTCCAAAAG-3′; human Snail, forward, 5′- ACTGCGACAAGGAGTACACC, and reverse, 5′- GAGTGCGTTTGCAGATGGG human Twist, forward, 5′- GCCTAGAGTTGCCGACTTATG-3′,and reverse, 5′- TGCGTTTCCTGTTAAGGTAGC-3′.

2.4. Cytokine analysis by ELISA

The expression levels of IL-1β, IL-6 and TNF-α from cell culture supernatants and mice colon tissue were analyzed by ELISA kits (R&D).

2.5. Flow cytometry analysis for PBMC from IBD patients

Peripheral blood collected from 20 CD patients (CDAI score > 180) and facilitated density gradient centrifugation to obtaine PBMC. PBMC from different patients were cultivated in 96-well plates with the density of 1×10^6/ml in DMEM which included 10% FBS, 100 U/mL penicillin, 100 μg/mL streptomycin. All samples were treated with ART in different concentration for 48 h and then harvested as single cell in PBS. The single-cell suspensions were stopped with 1%BSA and then stained by BV421-anti-CD11b and PE-anti-CD206 antibodies. Flow cytometry analysis was performed by Guava, and data were record and analyzed by FlowJo V10 software.

2.6. Organ cultures

Colon specimens were taken during endoscopy from the inflamed colon of CD patients. Each sample was cut into small pieces and divided into vehicle group and ART (100 mM) group. Samples were cultured for 6 h following the previous method [27]. Then, total RNA from each sample was extracted for qPCR analysis.

2.7. Western blot

Cells and tissues were collected in RIPA buffer. Proteins extract was obtained from crude membrane block then loaded on an SDS-PAGE with 8% polyacrylamide to research the expression level of ERK, pERK, MyD88, Arg-1, iNOS, Snail, Twist, occludin, claudin-1, ZO-1.

2.8. Histological processing and analysis

Extracted mouse intestines and made paraffin sections to provide compositions for histological analysis. The scores of colitis were graded in some part with hematoxylin and eosin (H&E) by two independent researchers followed the method reported previously [28]. Immunohistochemical (IHC) staining was performed on the representative areas using the markers F4/80, CD206 and MPO.

3. Result

3.1. ART treatment inhibited the secretion of inflammatory cytokines through inhibiting ERK signaling pathway in LPS stimulated RAW264.7 cells

LPS-stimulated RAW 264.7 cells were used to imitation M1 phenotype to evaluate the anti-inflammatory influence in ART *vitro*. Many pro-inflammatory cytokines illustrated as IL-1β, IL-6, TNF-α, etc. produced by M1 phenotype hold an essential role in the pathogenesis of IBD. To investigate the effect of ART on the secretion of the inflammatory mediators mentioned above at different levels, RT-PCR and ELISA analysis were performed. Fig. 1A-C showed that the gene expression of inflammatory cytokines was inhibited by ART treatment. Meanwhile, Fig. 1D-E show that the concentrations of IL-6 and TNF-α in the cell culture supernatants was down-regulated by ART treatment. To determine the effective of ART ameliorated M2-polarization of macrophages, the protein levels of iNOS and Arg-1 have detected by western blot and figure F shows the status of iNOS and Arg-1 were down regulated and up regulated respectively after ART treatment. Meanwhile, LPS stimulation led to an upregulation of ERK phosphorylation level, which can be inhibited ART, suggesting that ART blocked the secretions of inflammatory factors caused by LPS through inhibiting ERK phosphorylation.

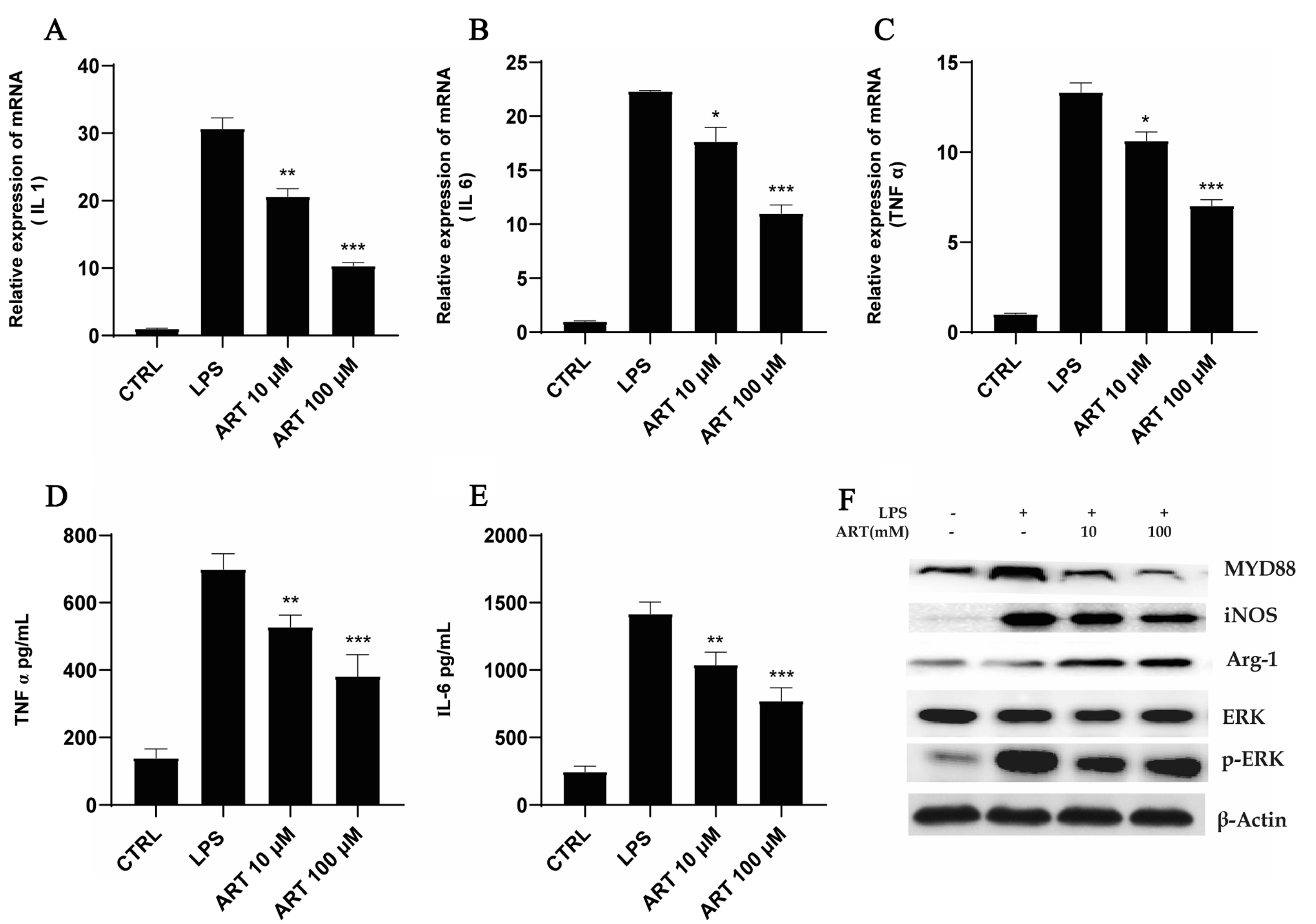

Fig. 1. ART treatment inhibited the secretion of inflammatory cytokines in LPS-stimulated RAW 264.7 cells. LPS-stimulated RAW 264.7 cells were treated by ART and the gene expression of IL-1β, IL-6 was detected by qPCR (A-C). The cells were cultured for 24 h and the secretions of IL-6 (D) and TNF-α (E) were examined by ELISA in the cell culture supernatants. The cells were cultured for 24 h and the expression of Arg-1, iNOS MYD88 and ERK was examined by Western blot (F). Data are record and analyzed as mean ± SEM. *P < 0.05, ** P < 0.01, ***P < 0.001 *vs* DSS group.

3.2. ART attenuated colitis in DSS-induced mice model

DSS-induced mice experiment colitis model was frequently used in IBD research, and the characters of the colitis model include weight loss, diarrhea and colitis bleeding [29]. As mentioned in Fig. 2A, mice was experienced severely weight loss, which attenuated by ART in another two groups. ART also reduced the increase of DAI score caused by DSS (Fig. 2B). The shortening of mice colon was recovered after ART intervention compared with vehicle control group (Fig. 2C and 2E). Meanwhile, the increasing of spleen index induced by DSS was also reversed by 80 mg/kg ART administration (Fig. 2D and F). The severity of colitis was analyzed by H&E staining of Intestinal paraffin section and histology score (Figure G and H). The results showed that the villi structure has been significantly repaired while less residual inflammatory cell infiltration in ART-treated mice.

3.3. ART regulating the cytokine network in DSS-induced colitis by skewing macrophages into M2 phenotype

Macrophages is of great importance in intestinal mucosal immunity. The main function of M1 subtype of macrophages is the secretion of inflammatory factors and mediates tissue damage as the main biological function, and the M2 subtype is involved in injury repair. In the state of intestinal immune imbalance, dysregulation of intestinal immune response will lead to abnormal regulation of inflammatory cytokine network, which may be manifested by increased secretions of various cytokines and related to inflammation activation [25]. We first detected the effect on the expression levels of pro-inflammatory mediators by ART in mice colon by ELISA. As shown in Fig. 3A, the expression levels of IL-1β, IL-6 and TNF-α were obviously increased in DSS-induced colitis mice. Notably, ART treatment significantly inhibited the expression of IL-1β, IL-6 and TNF-α in the inflamed intestine. We detected in the expression level of MPO in mice colon by IHC. Compared with the solvent control group, ART intervention obviously inhibited the increase of MPO expression caused by DSS (Fig. 3B). We detected the expression of F4/80 and CD206 by IHC in mice colon of each group. Fig. 3B showed theF4/80 expression in the mouse colon was significantly up-regulated in DSS induced colitis, and ART intervention significantly inhibited F4/80 expression and up-regulated CD206 expression.

3.4. ART ameliorates experimental colitis by inhibiting EMT process

EMT is related to the fibrosis of the intestinal tract and the injury of the intestinal epithelial barrier, which can cause colon shortening and increased permeability of the intestinal mucosa, leading to disorders of mucosal immune homeostasis. In vivo, the levels of EMT-related markers in the intestines of mice in each group were detected by qPCR. We found the expressions of EMT markers such as Snail, Twist, Fibronectin and Vimentin were up-regulated in DSS solvent-controlled mice, while the expression of epithelial cell markers which include ZO-1, Claudin-1 and Occludin were down-regulated(Fig. 4A), indicating that EMT is involved in the development of DSS-induced colitis. The treatment of ART reversed the up-regulation of Snail and Twist, suppressed the up-regulation of Fibronectin, and restored the performance levels of ZO-1, Claudin-1 and Occludin to a certain extent (Fig. 4B).

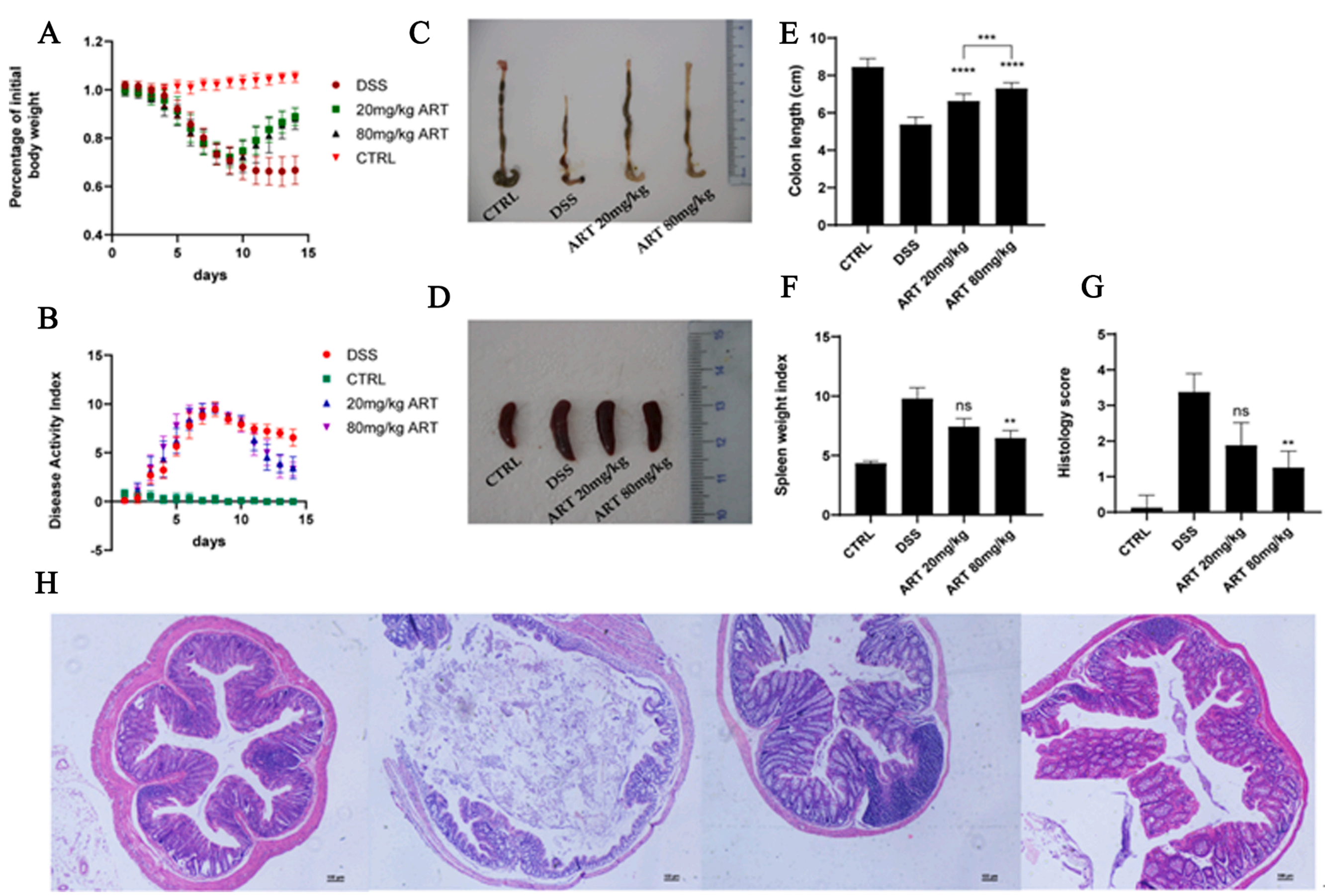

Fig. 2. ART treatment ameliorated colitis in DSS-induced mice model. Decrease of initial weight in all group of mice during the experience (A). DAI evaluated in each group of mice (B). Coolon length in each group of mice (C and E). Weight data in each group of mice (D and F). H&E stained colonic composition (H). Scores in colonic composition in each group mice (G). Data were put forward as means ± SEM (n = 8).*P < 0.05, ** P < 0.01, ***P < 0.001, **** P < 0.0001 *vs* DSS group.

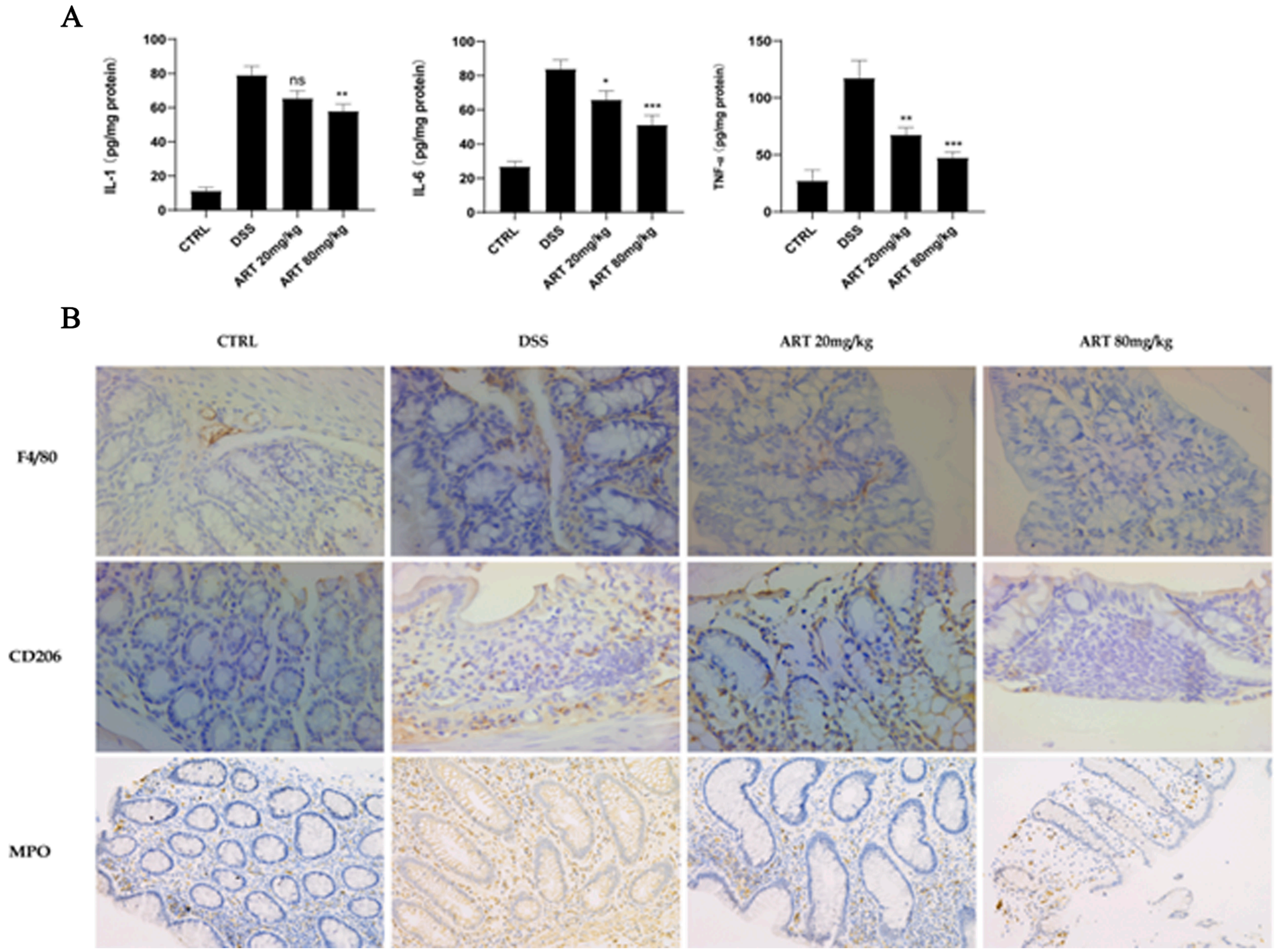

Fig. 3. ART regulating the cytokine network in DSS-induced colitis by skewing macrophages into M2 phenotype. The levels of cytokines (IL-1β, IL-6, TNF-α) in mice colonic tissue (A). Immunohistochemical staining of F4/80, CD206 and MPO in mice colonic tissues (B).

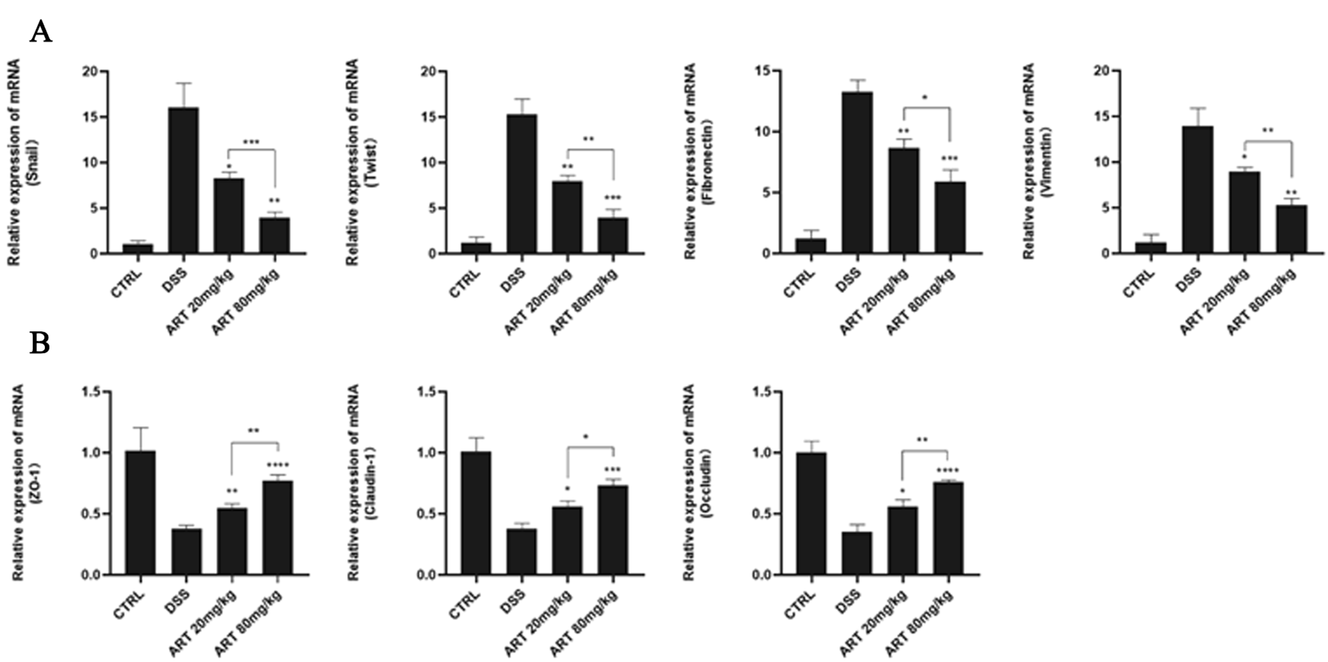

Fig. 4. ART ameliorates experimental colitis by inhibiting EMT. The mRNA expression of Snail, Twist, Fibronectin, Vimentin (A) and ZO-1, Claudin-1 and Occludin (B) in colonic tissues. Data were analyzed as means ± SEM (n = 6). *P < 0.05, ** P < 0.01, ***P < 0.001, **** P < 0.0001 *vs* DSS group.

3.5. ART inhibits EMT and regulats the polarization of macrophages through inhibiting the MYD88 and ERK pathway

To further prove the therapeutic effect of ART on colitis and explore its mechanism, the protein levels about macrophages polarization-related markers, EMT-related markers and related signaling pathways in DSS induced colitis mice model were investigated. Western blot analysis confirmed that ART down streamed the expression level of EMT related transcription factors Snail and Twist in contrast with the solvent control group in DSS mice model (Fig. 5A). Also, the expression levels of epithelial cell markers, ZO-1, Occludin and Cluadin-1, were up-regulated by ART in colitis mice. In addition, the expression level of Arg-1 was up-regulated while the performance level of iNOS was suppressed by ART in mice colitis model (Fig. 5A). Meanwhile, ART significantly suppressed MYD88 activation and ERK phosphorylation in colon tissue from DSS induced mice model (Fig. 5B). These results suggested that ART complied the effect of inhibiting EMT and regulating the polarization of macrophages through inhibiting the MYD88 and ERK pathway in DSS induced mice model.

3.6. ART has anti-inflammatory effects on human tissue samples

To expand our study, PBMC and colitis samples were collected from CD patients and treated with ART. To verify the anti-inflammatory effect and EMT inhibitory effect of ART in human samples, peripheral blood of patients with active CD was taken (n = 20, CDAI > 180), and PBMC were separated from peripheral blood and cultured in vitro for 48 h. FACs detection showed that ART induced macrophages to polarize to the M2 subtype and significantly increased the proportion of CD11b+CD206+ macrophages increased significantly (Fig. 6A and B). The pro-inflammatory factor concentration in the cell culture supernatants was tested by ELISA method, and it was found that the secretion of TNF-α and IL-6 of PBMC was significantly inhibited when the concentration of ART was 100 μM (Fig. 6C). Colon samples from patients with active CD were also treated by ART in vitro. We found that ART intervention down-regulated the gene expression levels in promote the inflammatory factors with EMT-related transcription factor Snail and Twist in the inflammatory intestinal tissue (Fig. 6D).

4. Discussion

As a kind of self-destructive gastrointestinal tract disease, IBD is described as a dysfunctional immune response to the intestinal-derived antigens. In a long period of time in history, IBD was a common sickness in affluent European and American countries, however, this incidence and prevalence of IBD increasing sharply in a series of Asian countries with the socioeconomic development [30] which has given a heavy impact on medical system and individual health. At present, due to the poor efficacy, side effects and expensive price, the clinical application of classical medicine (steroids, azathioprine, methotrexate, mesalazine, etc.) and biological agents is subject to many restrictions. Thus, a new affordable therapeutic agent with better efficacy and safety meets the clinical needs. Our study aimed at investigating the anti-inflammation effect of ART in LPS-stimulated RAW264.7 cells, DSS-induced experiment colitis mice and tissue from IBD patients. Our study showed that ART significantly ameliorated DSS-induced colitis by skewing macrophages to M2 phenotype and inhibiting EMT process. In addition, ART also inhibit the levels of pro-inflammatory cytokines though the suppression of ERK signaling pathway. The anti-inflammatory effects of ART seems to show a dose-dependent manner *in vivo*, for the colon length and spleen weight index in the group of low dose of ART (20 mg/kg) was not restore compared with the vehicle group while the high dose of ART (80 mg/kg) received significant improvment. These results showed that ART could be a new effective agent for clinical treatment of IBD.

Previous studies have shown that various artemisinin analogs ameliorate experimental colitis by down-regulating the TLR4 receptor and inhibiting the activation of M1 phenotype macrophages [28,31], which confirmed the importance of innate immunity for the development and outcome for IBD. In this study, ART was administered orally to evaluate its protective effect on the acute colitis induced by DSS. Our results show that ART has significantly improved acute colitis induced by DSS through increasing body weight, decreasing intestinal bleeding and mitigating colon shortening. iNOS and Arg-1 were signed to active M1 and M2 phenotype respectively, which be of great importance in the biological function of MΦ, respectively [32,33]. In our study, we stimulated RAW264.7 cells with LPS to mimic inflammation-activated M1 subtype macrophages and found that iNOS expression was up-regulated. ART intervention can suppress the up-regulation of iNOS expression caused by LPS and up-regulate the expression of Arg-1. Also, in contrast with the DSS control group, the level of iNOS was up-regulated and Arg-1 was up-regulated after ART treatment. PBMC from IBD patients also treated by ART and we also found that ART promote the M2 phenotype polarization. These results showed that the therapeutic effect of ART on DSS induced colon injury is related to the promotion of mononuclear cells in peripheral blood to M2 phenotype.

Pro-inflammatory cytokines not only holds an essential part in pathogenesis of DSS-induced colitis and IBD patients, their level also

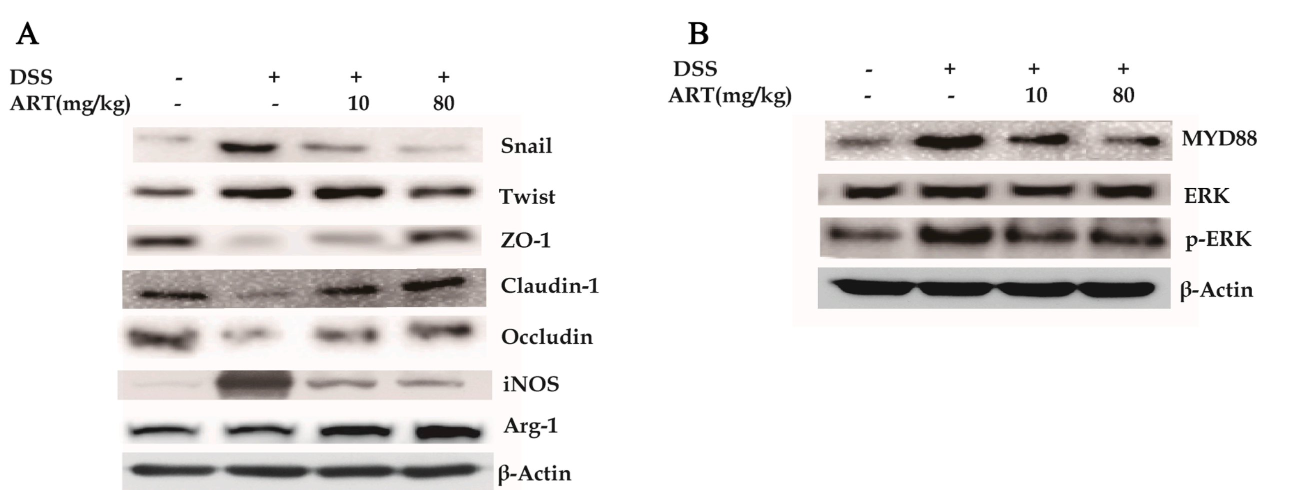

Fig. 5. ART regulated EMT process, macrophages polarization and related signaling pathways in DSS-induced colitis mice. Immunoblot analysis of the level of Snail, Twist, ZO-1, Occludin and Claudin-1 in the colonic paraffin section from each group (A). Immunoblot analysis of the level of MYD88, ERK, pERK, Arg-1, iNOS in the colonic paraffin section from each group (B).

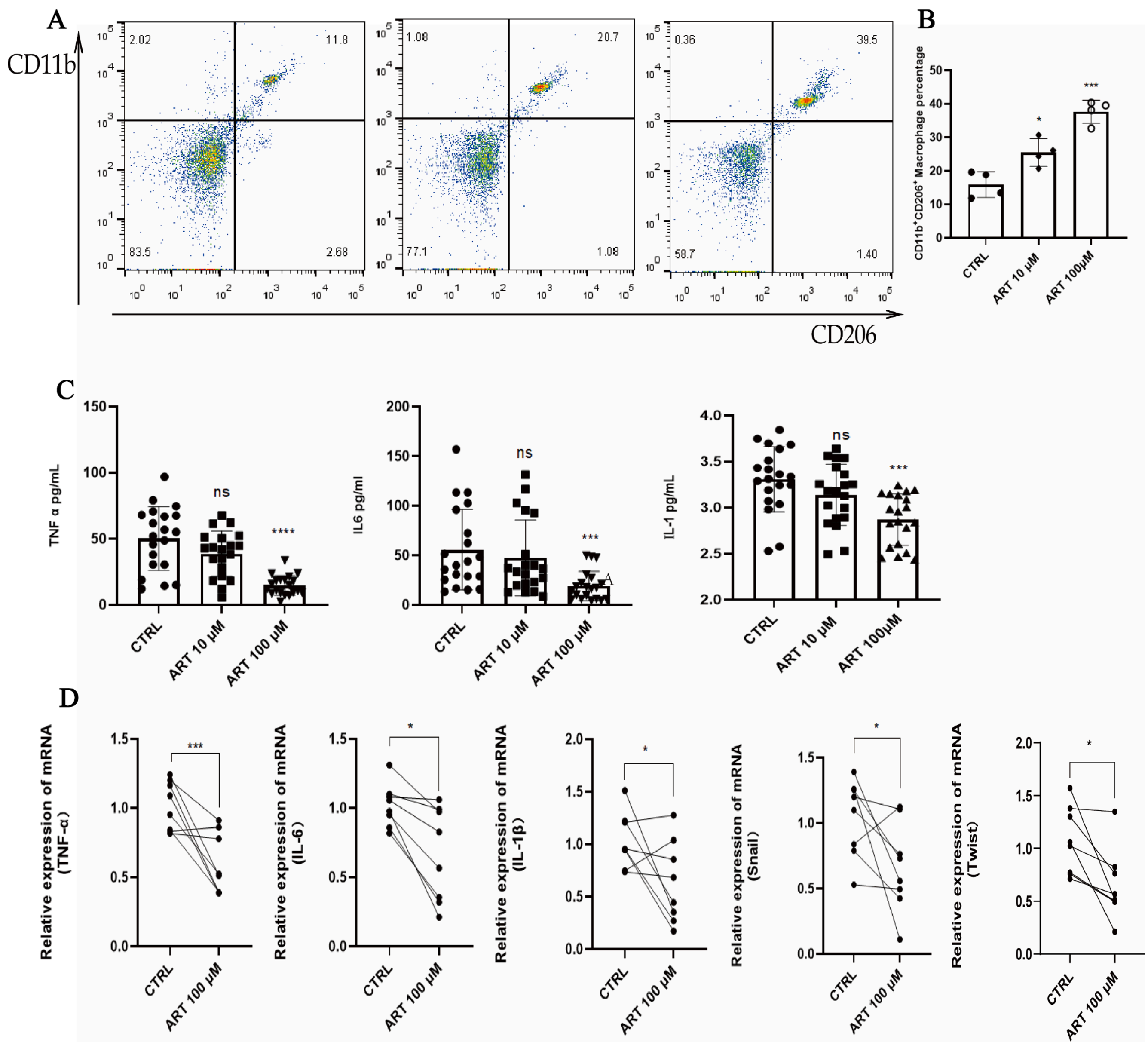

Fig. 6. ART showed anti-inflammatory effect and EMT inhibitory effect in human samples. PBMC and colon samples from active CD patients were incubate in vitro and cured with the indicated density of ART for 48 h (A-C) & 6 h (D), respectively. The percentage of CD11b+CD206+ macrophages from PBMC were measured by FACS analysis (A and B). The compositionof IL-1β, IL-6 and TNF-α were eaxmed by ELISA in supernatant of cell (B). The gene expression of IL-1β, IL-6, TNF-α, Snail and twist with colon tissue were eaxmed by qPCR (D). Data were studied by the *t*-test. *P < 0.05, ** P < 0.01, ***P < 0.001, **** P < 0.0001 *vs* DSS group. *vs* solvent control group.

correlate with the severity of disease [34,35]. Pro-inflammatory factors are also important aims for various immune system related disorders such as IBD, Spondyloarthritis, psoriasis, etc., and a variety of biological preparations already include TNF-α antagonists [36], IL-6 antagonists [37], IL-17 [38] antagonists have achieved great clinical efficacy. The current study suggested that ART significantly suppressed both mRNA and protein levels in IL-1β, IL-6, TNF-α in the mice experiment colitis model. Our results confirmed that ART administration markedly reversed the increased level of inflammation in mice colon by inhibiting the expression levels of pro-inflammatory cytokines, meanwhile include IL-1β, IL-6 & TNF-α. Taken together, these data further indicated that ART suppressed M1 polarization and promoted M2 polarization, leading to the down-streamed of pro-inflammatory mediators and iNOS.

In addition to intestinal immune dysfunction, intestinal epithelial barrier function damage caused by EMT is also one of the causes of IBD. In the process with EMT, epithelial cells lost these polarity, and the expression of epithelial cell signs ZO-1, Claudin-1 and Occludin are down-regulated, resulting in damage to the close linked between intestinal epithelial cells and affecting protection function. In animal experiments, we confirmed that the expressions of EMT-related transcription factors Snail and Twist in the intestinal tract of mice in the DSS group were up-regulated, and the performances in ZO-1, Claudin-1 and Occludin had down-regulated. After ART treatment, the expression levels of Snail and Twist were suppressed, and the performances levels of ZO-1, Claudin-1, Occludin were partially restored. This result indicates that ART can not only inhibit intestinal inflammation by inducing regulating the secretions in inflammatory factors, but also improve intestinal homeostasis by improving the inhibition of EMT and repairing the intestinal epithelial barrier.

MYD88 is a central player in innate immune signaling which recruited by TLR4. The activation of MYD88 leads to TNF receptor-associated factor 6 (TRAF6) activation, and eventually leads to the active signaling cascades of MAPKs [39,40]. It transfers extracellular signals into the cell and activates MAPK through the intracellular cascade phosphorylation. There are five groups of MAPKs have been defined in mammals and ERK pathway has been widely studied. ERK1/2 can be activated by cytokines and translocate into nucleus acting as a key regulator of cell differentiation and proliferation [19]. In our data, DSS induction of colitis showed a rapid phosphorylation of ERK in colonic tissue, which obviously suppressed by ART treatment. These results suggested that ART ameliorates the colitis in DSS -induced mice model by suppressing ERK activation.

To validate the potential efficacy of ART in patients with CD, we will start study the protective influence about ART in vitro using inflamed colon tissue and PBMC from active CD patients. ART administration importantly inhibited the secretions in PBMC from CD patients. In addition, we revealed that ART shows anti-inflammation effect on PBMC from CD patients by promoting macrophages polarized to M2 phenotype as evaluated by FACs analysis. In the in vitro culture of inflammatory intestinal tissue, ART obviously reduced mRNA level in pro-inflammatory factors and the intestinal tissue and down-regulated the transcription factor Snail and Twist mRNA level. These data proved that ART can reduce the inflammatory response of CD patients by promoting the polarization of macrophages towards M2 phenotype, blocking the performance about the pre-inflammatory factors and this process of EMT.

In a word, our study found a new treatment method for inflammatory bowel disease by regulating the polarization of macrophages and EMT process with ART. Our data proved that ART could improve DSS induced experimental colitis mainly by skewing macrophages polarizing to M2 phenotype, and inhibiting the process of EMT and activation of ERK cell signaling pathway. Furthermore, ART also down-streamed the expression of inflammatory cytokines on inflamed colon tissue and PBMC from active CD patients and promote the polarization of M2 phenotype. Our study results proved that ART can be regarded as a selection therapy for clinical treatment of IBD. However, the other animal models for IBD research, such as TNBS-induced colitis, DSS- induced chronic colitis, IL-$10^{-/-}$ induced colitis, etc. ought to evaluate the efficacy of ART therapy for IBD. Meanwhile, we were unable to obtain more samples from CD or UC patient due to various restrictions. In the follow-up work, we will collect more samples of CD and UC patients for experimental verification.

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CRediT authorship contribution statement

Manxiu Huai: Methodology, Data curation, Writing - original draft, Writing - review & editing. **Junxiang Zeng:** Data curation, Investigation, Writing - original draft, Writing - review & editing. **Wensong Ge:** Supervision, Funding acquisition, Methodology.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intimp.2020.107284.

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